

Atomic Mass, Percentage Composition, and Chemical Formulas

Every atom has mass, but that mass is incredibly tiny – too small to measure directly on a balance. A single carbon atom weighs about 0.000000000000000000000002 grams. So how do chemists know the mass of atoms? They use a relative scale, comparing every atom to a standard. This system of relative atomic masses allows chemists to count atoms by weighing them. Once the atomic masses are known, the composition of compounds can be calculated, and the formulas of unknown substances can be determined.

Atomic mass tells chemists how much one atom weighs compared to another. Percentage composition tells chemists what a compound is made of. Empirical and molecular formulas tell chemists the exact identity of a substance. This reading material explores the difference between relative atomic mass and average atomic mass, how to calculate percentage composition from a formula, the similarities and differences between empirical and molecular formulas, and how to calculate empirical and molecular formulas from experimental data.

Part 1: What Is Atomic Mass and Why Does It Matter?

Definition of Atomic Mass

Atomic mass is the mass of an atom, typically expressed in atomic mass units (u or amu). Because individual atoms are far too small to weigh directly, chemists use a relative scale based on a standard reference isotope.

The Central Idea: Atomic Mass Is Relative

You cannot weigh a single atom on a balance. Instead, chemists compare the mass of one atom to the mass of another atom. The standard is carbon-12 (^{12}C), which is

assigned an exact mass of 12 atomic mass units. All other atomic masses are measured relative to this standard.

Why Atomic Mass Matters

Aspect	Explanation
Counting atoms	Atomic mass allows chemists to count atoms by weighing samples (using moles)
Determining formulas	Atomic masses are used to calculate empirical and molecular formulas
Predicting reactions	Atomic masses are needed to balance equations and calculate yields
Identifying elements	Each element has a unique atomic mass (average of its isotopes)
Quality control	Atomic mass calculations verify the purity and composition of compounds

Part 2: Relative Atomic Mass vs. Average Atomic Mass

The terms "relative atomic mass" and "average atomic mass" are often used interchangeably, but there is a subtle and important difference. Understanding this difference is essential for accurate calculations.

Definitions

Term	Definition	Symbol	What It Represents
Relative atomic mass	The mass of an atom of an element compared to 1/12th the mass of a carbon-12 atom	A_r (or A_r)	The mass of one specific isotope relative to the standard

Average atomic mass	The weighted average of the masses of all naturally occurring isotopes of an element	Atomic weight (on periodic table)	The average mass of a typical sample of the element, considering isotope abundance
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Key Difference

Feature	Relative Atomic Mass	Average Atomic Mass
Refers to	A single specific isotope	A mixture of isotopes (natural sample)
Takes isotope abundance into account?	No	Yes
Is it the same for all samples of the element?	Yes (same isotope)	Slightly varies if isotope ratios change (rare)
Where is it found?	Listed for specific isotopes in nuclear data tables	Listed on the periodic table
Example for Carbon	Carbon-12 has relative atomic mass = 12.0000 exactly	Average atomic mass of carbon = 12.011 u (includes C-12, C-13, and trace C-14)

Why the Difference Matters

Importance	Explanation
Precision in calculations	Using the wrong value (isotopic vs. average) leads to incorrect results
Understanding isotopes	The difference between relative and average atomic mass highlights the existence of isotopes
Real-world samples	Most chemical samples contain natural mixtures of isotopes, so the average atomic mass is usually the correct value to use

Importance	Explanation
Mass spectrometry	Mass spectrometers measure relative atomic masses of individual isotopes; average atomic mass is calculated from these data

Calculating Average Atomic Mass from Isotopic Data

Average atomic mass is calculated using the formula:

$$\text{Average atomic mass} = \Sigma (\text{fractional abundance} \times \text{isotopic mass})$$

Where Σ means "sum of" and fractional abundance = percentage abundance $\div$ 100.

Example: Chlorine

Chlorine has two naturally occurring isotopes:

- Chlorine-35: mass = 34.9689 u, abundance = 75.78%
- Chlorine-37: mass = 36.9659 u, abundance = 24.22%

Calculation:

- Convert percentages to fractions: 75.78% = 0.7578; 24.22% = 0.2422
- Multiply each isotopic mass by its fraction: $(34.9689 \times 0.7578) = 26.50$ u
- $(36.9659 \times 0.2422) = 8.95$ u
- Add them together: 26.50 u + 8.95 u = 35.45 u

Result: The average atomic mass of chlorine is approximately 35.45 u, which matches the value on the periodic table.

Part 3: Percentage Composition of Compounds

Percentage composition is the percentage by mass of each element in a compound. It tells you what fraction of a compound's total mass comes from each element. For example, if water is 11.21% hydrogen and 88.79% oxygen by mass, then every 100 grams of water contains 11.21 grams of hydrogen and 88.79 grams of oxygen. This information is essential for identifying unknown compounds, verifying purity, and calculating empirical formulas.

Why Percentage Composition Is Important

Importance	Explanation
Identifying unknown compounds	The percentage composition of an unknown compound can be compared to known compounds
Verifying purity	The experimental percentage composition of a sample can be compared to the theoretical value
Determining formulas	Percentage composition data are used to calculate empirical formulas
Quality control	Manufacturers use percentage composition to ensure products meet specifications

How to Calculate Percentage Composition from a Formula

To calculate the percentage composition of a compound, follow these steps:

Step	Action	Example (CaCO ₃)
1	Identify the elements in the compound	Calcium (Ca), Carbon (C), Oxygen (O)
2	Determine the number of atoms of each element from the formula	Ca: 1 atom, C: 1 atom, O: 3 atoms
3	Find the atomic mass of each element (from periodic table)	Ca: 40.08 u, C: 12.01 u, O: 16.00 u
4	Calculate the total mass contributed by each element	Ca: $1 \times 40.08 = 40.08$ u C: $1 \times 12.01 = 12.01$ u O: $3 \times 16.00 = 48.00$ u
5	Calculate the molar mass (sum of all masses)	$40.08 + 12.01 + 48.00 = 100.09$ u
6	Divide each element's mass by the total mass and multiply by 100%	Ca: $(40.08 \div 100.09) \times 100\% = 40.04\%$ C: $(12.01 \div 100.09) \times 100\% = 12.00\%$ O: $(48.00 \div 100.09) \times 100\% = 47.96\%$

Step	Action	Example (CaCO ₃)
7	Check that the percentages add up to 100%	40.04% + 12.00% + 47.96% = 100.00%

Worked Example: Calcium Carbonate (CaCO₃)

Element	Number of Atoms	Atomic Mass (u)	Total Mass (u)	Percentage Composition
Calcium (Ca)	1	40.08	40.08	$(40.08 \div 100.09) \times 100\% = 40.04\%$
Carbon (C)	1	12.01	12.01	$(12.01 \div 100.09) \times 100\% = 12.00\%$
Oxygen (O)	3	16.00	48.00	$(48.00 \div 100.09) \times 100\% = 47.96\%$
Total			100.09	100.00%

Additional Worked Examples

Example 1: Water (H₂O)

Element	Number of Atoms	Atomic Mass (u)	Total Mass (u)	Percentage Composition
Hydrogen (H)	2	1.01	2.02	$(2.02 \div 18.02) \times 100\% = 11.21\%$
Oxygen (O)	1	16.00	16.00	$(16.00 \div 18.02) \times 100\% = 88.79\%$
Total			18.02	100.00%

Example 2: Glucose (C₆H₁₂O₆)

Element	Number of Atoms	Atomic Mass (u)	Total Mass (u)	Percentage Composition
Carbon (C)	6	12.01	72.06	$(72.06 \div 180.18) \times 100\% = 40.00\%$
Hydrogen (H)	12	1.01	12.12	$(12.12 \div 180.18) \times 100\% = 6.73\%$
Oxygen (O)	6	16.00	96.00	$(96.00 \div 180.18) \times 100\% = 53.27\%$
Total			180.18	100.00%

Part 4: Empirical Formula vs. Molecular Formula

Chemists use two types of chemical formulas to describe compounds: the empirical formula and the molecular formula. Each serves a different purpose.

Definitions

Term	Definition	Example (Hydrogen Peroxide)
Empirical formula	The simplest whole-number ratio of atoms of each element in a compound	HO (1:1 ratio of H to O)
Molecular formula	The actual number of atoms of each element in one molecule of the compound	H ₂ O ₂ (2 H atoms and 2 O atoms per molecule)

Importance of the Empirical Formula

The empirical formula gives the simplest whole-number ratio of atoms in a compound. It is important because it serves as the starting point for identifying unknown substances

and calculating the molecular formula. once the empirical formula and the molar mass are known, the true molecular formula can be calculated.

Importance of the Molecular Formula

The molecular formula gives the actual number of atoms of each element in one molecule of a compound. It is important because it uniquely identifies a specific compound, whereas different compounds can share the same empirical formula. For example, glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) and acetic acid ($\text{C}_2\text{H}_4\text{O}_2$) both have the same empirical formula (CH_2O) but are completely different substances. Only the molecular formula tells you exactly which compound you have.

Comparison Table: Empirical Formula vs. Molecular Formula

Feature	Empirical Formula	Molecular Formula
What it shows	Simplest ratio of atoms	Actual number of atoms per molecule
Can it be the same as the molecular formula?	Yes (if the compound is in simplest ratio)	Yes (if the compound is in simplest ratio)
Example (Glucose)	CH_2O	$\text{C}_6\text{H}_{12}\text{O}_6$
Example (Water)	H_2O	H_2O (same)
Example (Ethylene)	CH_2	C_2H_4
Example (Benzene)	CH	C_6H_6
How is it determined?	From percentage composition data	From empirical formula AND molar mass
Is it unique to one compound?	No (different compounds can share the same empirical formula)	Yes (each compound has a unique molecular formula)

Relationship Between Empirical and Molecular Formulas

The molecular formula is always a whole-number multiple of the empirical formula.

$$\text{Molecular formula} = (\text{Empirical formula}) \times n$$

Where $n = (\text{molar mass of compound}) \div (\text{molar mass of empirical formula})$

Compound	Empirical Formula	Empirical Formula Mass	Molar Mass	n	Molecular Formula
Glucose	CH ₂ O	$12.01 + (2 \times 1.01) + 16.00 = 30.03 \text{ u}$	180.16 u	$180.16 \div 30.03 = 6$	C ₆ H ₁₂ O ₆
Ethylene	CH ₂	$12.01 + (2 \times 1.01) = 14.03 \text{ u}$	28.06 u	$28.06 \div 14.03 = 2$	C ₂ H ₄
Benzene	CH	$12.01 + 1.01 = 13.02 \text{ u}$	78.12 u	$78.12 \div 13.02 = 6$	C ₆ H ₆

Compounds That Share the Same Empirical Formula

Different compounds can have the same empirical formula but different molecular formulas.

Empirical Formula	Possible Molecular Formulas	Compound Names
CH ₂	C ₂ H ₄ , C ₃ H ₆ , C ₄ H ₈ , C ₅ H ₁₀	Ethylene, Propylene, Butene, Pentene
CH ₂ O	CH ₂ O, C ₂ H ₄ O ₂ , C ₃ H ₆ O ₃ , C ₆ H ₁₂ O ₆	Formaldehyde, Acetic acid, Lactic acid, Glucose
NH ₂	NH ₂ , N ₂ H ₄	Amidogen, Hydrazine

Part 5: Calculating Empirical Formula from Percentage Composition

To determine the empirical formula of a compound from percentage composition data, follow these steps:

Step-by-Step Procedure

Step	Action	Example (Unknown compound: 40.0% C, 6.7% H, 53.3% O)
1	Assume a 100 g sample (percentages become grams)	C: 40.0 g, H: 6.7 g, O: 53.3 g
2	Convert grams to moles using atomic masses	C: $40.0 \div 12.01 = 3.33$ mol H: $6.7 \div 1.01 = 6.63$ mol O: $53.3 \div 16.00 = 3.33$ mol
3	Divide each mole value by the smallest mole value	Smallest = 3.33 C: $3.33 \div 3.33 = 1.00$ H: $6.63 \div 3.33 = 1.99 \approx 2$ O: $3.33 \div 3.33 = 1.00$
4	Round to the nearest whole number (if close)	C = 1, H = 2, O = 1
5	Write the empirical formula using these ratios	CH ₂ O

Worked Example 1: Finding Empirical Formula

Problem: A compound is found to contain 92.3% carbon and 7.7% hydrogen. Determine its empirical formula.

Step	Calculation
1	Assume 100 g: C = 92.3 g, H = 7.7 g
2	Convert to moles: C = $92.3 \div 12.01 = 7.69$ mol; H = $7.7 \div 1.01 = 7.62$ mol
3	Divide by smallest (7.62): C = $7.69 \div 7.62 = 1.01 \approx 1$; H = $7.62 \div 7.62 = 1$
4	Empirical formula = CH

Answer: The empirical formula is CH.

Worked Example 2: When Ratios Are Not Whole Numbers

Problem: A compound contains 25.9% nitrogen and 74.1% oxygen. Find the empirical formula.

Step	Calculation
1	Assume 100 g: N = 25.9 g, O = 74.1 g
2	Convert to moles: N = $25.9 \div 14.01 = 1.85$ mol; O = $74.1 \div 16.00 = 4.63$ mol
3	Divide by smallest (1.85): N = $1.85 \div 1.85 = 1.00$; O = $4.63 \div 1.85 = 2.50$
4	Multiply by 2 to get whole numbers: N = $1 \times 2 = 2$; O = $2.50 \times 2 = 5$
5	Empirical formula = N ₂ O ₅

Answer: The empirical formula is N₂O₅ (dinitrogen pentoxide).

When to Multiply to Get Whole Numbers

Decimal Ending	Multiply By	Example
.0 (already whole)	1	1.00
.25	4	$1.25 \times 4 = 5$
.33 or .67	3	$1.33 \times 3 = 4$
.5	2	$1.50 \times 2 = 3$
.75	4	$1.75 \times 4 = 7$

Part 6: Calculating Molecular Formula from Empirical Formula and Molar Mass

Once the empirical formula is known, the molecular formula can be determined if the molar mass of the compound is also known.

Step-by-Step Procedure

Step	Action	Example (Empirical formula = CH_2O , molar mass = 180.16 g/mol)
1	Calculate the empirical formula mass	C: 12.01, H: $2 \times 1.01 = 2.02$, O: 16.00 Total = $12.01 + 2.02 + 16.00 = 30.03$ g/mol
2	Divide the given molar mass by the empirical formula mass	$n = 180.16 \div 30.03 = 6.00$
3	Multiply the empirical formula by n	$(\text{CH}_2\text{O}) \times 6 = \text{C}_6\text{H}_{12}\text{O}_6$
4	Write the molecular formula	$\text{C}_6\text{H}_{12}\text{O}_6$ (glucose)

Worked Example 1: Finding Molecular Formula

Problem: A compound has an empirical formula of CH_2 and a molar mass of 84.18 g/mol. Determine the molecular formula.

Step	Calculation
1	Empirical formula mass = $12.01 + (2 \times 1.01) = 12.01 + 2.02 = 14.03 \text{ g/mol}$
2	$n = 84.18 \div 14.03 = 6.00$
3	Molecular formula = $(\text{CH}_2) \times 6 = \text{C}_6\text{H}_{12}$
4	Answer: C_6H_{12} (cyclohexane or hexene)

Worked Example 2: Finding Molecular Formula with Non-Integer n

Problem: A compound has an empirical formula of NO_2 and a molar mass of 92.02 g/mol. Determine the molecular formula.

Step	Calculation
1	Empirical formula mass = $14.01 + (2 \times 16.00) = 14.01 + 32.00 = 46.01 \text{ g/mol}$
2	$n = 92.02 \div 46.01 = 2.00$
3	Molecular formula = $(\text{NO}_2) \times 2 = \text{N}_2\text{O}_4$
4	Answer: N_2O_4 (dinitrogen tetroxide)

Key Terms

Term	Definition
Relative atomic mass	The mass of an atom of a specific isotope compared to 1/12th the mass of carbon-12
Average atomic mass	The weighted average of the masses of all naturally occurring isotopes of an element
Atomic mass unit (u)	A unit of mass equal to 1/12th the mass of a carbon-12 atom
Percentage composition	The percentage by mass of each element in a compound
Empirical formula	The simplest whole-number ratio of atoms of each element in a compound
Molecular formula	The actual number of atoms of each element in one molecule of a compound
Molar mass	The mass of one mole of a substance (in g/mol), numerically equal to molecular or formula mass
Isotopic abundance	The percentage of a given isotope in a naturally occurring sample of an element

Review Questions

1. Explain the difference between relative atomic mass and average atomic mass. Your answer must include what each term refers to, whether isotope abundance is considered, and which value appears on the periodic table.

2. Chlorine has two naturally occurring isotopes: Chlorine-35 (mass = 34.9689 u, abundance = 75.78%) and Chlorine-37 (mass = 36.9659 u, abundance = 24.22%). Calculate the average atomic mass of chlorine. Show your calculations.

3. Calculate the percentage composition of calcium carbonate (CaCO_3). Your answer must include the percentage of calcium, carbon, and oxygen. Show all steps.

4. Calculate the percentage composition of the following compounds:

- (a) Water (H_2O)
- (b) Carbon dioxide (CO_2)
- (c) Sodium chloride (NaCl)

5. Complete the following table comparing empirical and molecular formulas:

Feature	Empirical Formula	Molecular Formula
What it shows		
Example for glucose		
Can different compounds have the same formula?		
How is it determined?		

6. A compound is found to contain 52.2% carbon, 13.0% hydrogen, and 34.8% oxygen by mass. Calculate the empirical formula of this compound. Show all steps.

7. A compound has the empirical formula CH_2O and a molar mass of 180.16 g/mol. Calculate the molecular formula. Show your calculations.

8. A compound contains 40.0% carbon, 6.7% hydrogen, and 53.3% oxygen. Its molar mass is 180.16 g/mol.

- (a) Determine the empirical formula.
- (b) Determine the molecular formula.
- (c) Identify the compound (name it).

9. Explain why the average atomic mass of carbon is 12.01 u even though carbon-12 has a relative atomic mass of exactly 12 u. Use the concept of isotopes in your answer.

10. Application question: A student analyzes an unknown compound and finds that it contains 75.0% carbon and 25.0% hydrogen. The molar mass is found to be 32.00 g/mol. Determine both the empirical formula and the molecular formula. Show all steps and name the compound if possible.